

No. of Functional Features Included in the Solution

Date	03 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Collects soil and environmental parameters from users	Frontend	Done	High
2	Input Validation	Validates all input values before prediction	Validation Module	Done	Medium
3	Data Preprocessing	Cleans and formats input data	Preprocessing	Done	High
4	Crop Prediction	Predicts the best crop using the trained ML model	ML Model	Done	High
5	Prediction Result Display	Result Display Displays recommended crop with confidence	Results Module	Done	Medium
6	Model Loading	Loads trained Random Forest model during application startup	Backend	Done	Medium
7	Error Handling	Handling Handles invalid inputs and runtime exceptions	Backend	Done	Medium
8	Responsive User Interface	Provides a simple and user-friendly interface	Frontend	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5

Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input Form, Result Display
2	Backend / Logic	3	Crop Prediction, Input Validation, Error Handling
3	Database / Storage	1	Trained Model (.pkl) and Dataset Storage
4	API / Integration	1	Flask Backend API for Prediction
5	Security / Authentication	1	

Input Validation and Error Handling